

1. What are the possible orders for the elements of S_6 and A_6 ?

$S_6 \rightarrow$ permutations of $\{1, \dots, 6\}$
 $-1, 2, 3, 4, 5, 6$

$A_6 \rightarrow$ even permutations of $\{1, \dots, 6\}$
 $-1, 2, 3, 4, 5$

2. Suppose α is a 6-cycle and β is a 5-cycle. Determine whether $\alpha^5 \beta^4 \alpha^{-1} \beta^{-3} \alpha^5$ is even or odd.

α is odd $\Rightarrow \alpha^5$ is odd, α^{-1} is odd
 β is even $\Rightarrow \beta^4$ is even, β^3 is even $\Rightarrow \beta^{-3}$ is even

$\alpha^5 \quad \beta^4 \quad \alpha^{-1} \quad \beta^{-3} \quad \alpha^5$
 $\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $O \quad E \quad O \quad E \quad O$
 $\Rightarrow$ odd.

3. Let $\alpha = (1, 3, 5, 7, 9)(2, 4, 6)(8, 10)$. If α^m is a 5-cycle, what can you say about m ?

$$|\alpha| = \text{lcm}(5, 3, 2) = 30$$

$$|\alpha^m| = 5 \quad (\alpha^m)^5 = \epsilon \Rightarrow 5m \text{ is a multiple of } 30$$

$$5m = 30n$$

$$m = 6n \Rightarrow m \text{ is a multiple of } 6.$$

4. Prove that (1234) is not the product of 3-cycles.

(1234) is an odd permutation

A 3-cycle is an even permutation

Thus three 3-cycles will also be even.

5. Let $\alpha_1, \alpha_2, \alpha_3$ be 2-cycles. Prove that $\alpha_1 \alpha_2 \alpha_3 \neq \epsilon$.

ϵ is even (it is in A_n)

but $\alpha_1 \alpha_2 \alpha_3$ is odd.

Thus they cannot be equal.

6. What are the orders of the elements of D_{15} ? How many elements have these orders?

• R_0 has order 1

• each flip (15) have order 2.

• To determine the other orders, we need to consider the other

rotations. Observe the set of rotations

- order 15: $\phi(15) = 8$

- order 5: $\phi(5) = 4$

- order 3: $\phi(3) = 2$

7. Suppose G is a group with exactly eight elements of order 10. How many cyclic subgroups of order 10 does G have?

let $a \in G$ s.t. $|a| = 10$. Then $\langle a \rangle$ is a cyclic subgroup of order 10. Observe $\langle a \rangle$ has $\phi(10) = 4$ elements of order 10, a, a^3, a^7 and a^9 . Thus each cyclic group of order 10 contains 4 elements of order 10.

Since there are 8 elements of order 10, there are $\frac{8}{4} = 2$ cyclic subgroups of order 10.

8. Prove that a group of order 4 must be Abelian.

Proof: If G has an element of order 4, G is cyclic & thus Abelian. Suppose G has an element of order 2, say a . Then $G = \{e, a, a^2, b\}$. Consider element ab . Then, $ab \neq e, ab \neq a, ab \neq a^2, ab \neq b$, by the cancellation laws. If all non-identity elements have order 2, then $a, b \in G$ w/ $|a| = |b| = 2$. Consider $(ab)^2 = (ab)(ab) = e$. Also, $a^2b^2 = e \neq abab$. Thus $abab = abab$. And by cancellation $ab = ba$. Thus G is Abelian.

9. The torsion subgroup is the subset of elements of finite order in an Abelian group G . Prove that this is a subgroup of G . Let T be the torsion subgroup.

Two-step subgroup test.

• nonempty: since $|e| = 1, e \in T$.

• closure: let $a, b \in T$. Then $|a| = n, |b| = m$ for some finite $m, n \in \mathbb{Z}$. let $p = \text{lcm}(m, n)$. Then $m|p + n|p$. Now consider $(ab)^p = a^p b^p = (a^m)^{\frac{p}{m}} (b^n)^{\frac{p}{n}} = e \cdot e = e$. Thus $|ab| \leq p$ and $|ab|$ is finite.

• inverse: we have proven that $|a| = |a^{-1}|$. Thus if a has finite order, so does a^{-1} .

Thus by 2-step subgroup test, T is a subgroup of G .

10. Suppose G is a group and G has exactly two non-trivial proper subgroups. Prove that G is cyclic and $|G| = pq$, where p and q are distinct primes, or that G is cyclic and $|G| = p^3$, where p is a prime.

Proof: Let G be a group w/ exactly 2 non-trivial subgroups.
obtain one of these proper subgroups is $\{e\}$.

① [Show G is cyclic]: Spoke for contradiction that G is not

cyclic. Then $\exists x \in G$ w/ $\langle x \rangle \neq G$. Thus $\langle x \rangle$ is

the non-identity proper subgroup of G . Since $\langle x \rangle$ is

proper, $\exists y \in G$ w/ $y \notin \langle x \rangle$. Since G has only 2 proper

subgroups, $G = \langle x \rangle \cup \langle y \rangle$. Since $xy \in \langle x \rangle$, $xy \in \langle y \rangle$.

② Don't worry about the order for now.

11. Let G be a cyclic group of order n and let H be the subgroup of G of order d . Show that $H = \{x \in G : |x| \text{ divides } d\}$.

Let $G = \langle a \rangle$ where $|a| = n$. Let $H \leq G$ s.t. $|H| = d$, where $n = ds$.
Then $H = \langle a^{n/d} \rangle = \langle a^s \rangle$, by FT. CG.

Now let $K = \{x \in G : |x| \text{ divides } d\}$.

① Show $K \leq H$: Let $x \in K$. Then $|x| = d$, and $\exists t \in \mathbb{Z}$ where $|x| \cdot t = d \Rightarrow |x| \cdot t = \frac{n}{s} \Rightarrow |x| = \frac{n}{st}$.

By FT. CG, $\langle x \rangle = \langle a^{n/|x|} \rangle = \langle a^{n/(n/st)} \rangle = \langle a^{st} \rangle = \langle (a^s)^t \rangle$.
Since $\langle x \rangle = \langle (a^s)^t \rangle$, $\langle x \rangle \leq \langle a^s \rangle = H$ and $x \in H$.

Thus $K \leq H$.

② Show $H \leq K$: Let $h \in H$. Then $|h|$ divides $|H|$.
Thus $|h|$ divides d . By def of K , this

implies $h \in K$, and $H \leq K$.

Since $K \leq H + H \leq K$, $H = K$, and $H = \{x \in G : |x| \text{ divides } d\}$.

